UNLOCKING DETAILED IMAGE INTERPRETATION THROUGH SEMANTIC SEGMENTATION USING AI

A Project Report submitted for the partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

SUBMITTED
BY

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MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE

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MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE

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DEPARTMENT OF COMPUTER APPLICATIONS

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This is to certify that the project work entitled **"Unlocking Detailed Image Interpretation through semantic segmentation using AI"** is a bonafide work carried out by L Binni (Roll No.: 23691F0025), submitted in the partial fulfillment of the requirements for the award of the degree of Master of Computer Applications in Madanapalle Institute of Technology & Science, Madanapalle, affiliated to Jawaharlal Nehru Technological University Anantapur, Anantapuram the academic year 2024-2025.

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I, L Binni **(Roll No.: 23691F0025)** hereby declare that the project entitled **"Unlocking Detailed Image Interpretation through semantic segmentation using AI**" is done by me under the guidance of **Dr.C. SIVARAJ, MCA.,** submitted in partial fulfillment of the requirements for the award of degree of Master of Computer Applications at **MADANAPALLE INSTITUTE OF TECHNOLOGY & SCIENCE**, Madanapalle affiliated to Jawaharlal Nehru Technological University Anantapur, Anantapuram during the academic year 2024-2025. This work has not been submitted by anybody towards the award of any degree.

Date:

Place: Madanapalle

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First, I must thank the almighty who granted me knowledge, wisdom, strength, and courage to serve in this world and to carry out my project work in a successful way.

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L Binni

ABSTRACT

In today's data-driven world, extracting meaningful insights from visual content has become critical, especially in fields like healthcare, autonomous systems, urban planning, and surveillance. Traditional image analysis methods often fall short in capturing complex contextual information. To address this, our project leverages **semantic segmentation using AI** to enable fine-grained image understanding by assigning a semantic label to every pixel in an image.

This project utilizes advanced deep learning architectures such as **U-Net** and **DeepLabV3+**, trained on benchmark datasets like **PASCAL VOC** and **Cityscapes**, to segment various objects including roads, pedestrians, buildings, and vehicles. By focusing on pixel-level classification, the system enhances object boundary detection, contextual recognition, and spatial structure interpretation.

The trained models achieved competitive performance, with mean Intersection over Union (mIoU) scores reaching up to **76.4%** on validation datasets. Compared to standard classification or object detection, our segmentation model provides a more detailed and interpretable representation, making it ideal for real-world applications like automated driving, medical diagnostics, and smart surveillance.

The system has been tested on both static images and live video frames, demonstrating its real-time applicability and accuracy, thus offering a powerful tool for enhanced visual analysis through AI.

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LIST OF ABBREVAITIONS

S.NO.	ABBREVIATED TERM	EXPANSION
1	UML	Unified Modeling Language
2	ER Diagram	Entity Relationship Diagram

CHAPTER-I

INTRODUCTION

1.1 About The Project

Understanding visual content with high accuracy is a critical challenge in various domains such as autonomous vehicles, medical diagnostics, urban planning, and smart surveillance. Traditional image processing techniques often fall short in providing pixel-level detail, which is essential for comprehensive scene interpretation.

This project introduces an AI-powered solution that utilizes semantic segmentation to enable detailed pixel-wise image understanding. Unlike simple object detection or image classification, semantic segmentation assigns a class label to each pixel in the image, allowing for a richer understanding of the spatial layout and context.

Our approach is built upon state-of-the-art deep learning models such as **U-Net** and **DeepLabV3+**, trained on widely used datasets like PASCAL VOC and Cityscapes. These models are capable of segmenting complex scenes involving multiple objects, such as roads, pedestrians, vehicles, vegetation, and sky, with high accuracy.

By using encoder-decoder architectures and atrous convolutions, the models capture both global and local contextual information. This project aims to support real-time image analysis in smart applications, where detailed scene interpretation is critical—such as autonomous navigation, smart surveillance, and healthcare diagnostics.

With its strong performance and practical relevance, the system reduces manual efforts, enhances visual interpretation, and lays a foundation for more intelligent and context-aware computer vision systems.

CHAPTER II

SYSTEM ANALYSIS

2.1 Existing System

Traditional image analysis systems often rely on classification or object detection models that can identify objects within an image but lack the capability to understand the scene at a pixel level. These methods provide only bounding boxes and do not offer fine-grained detail, which limits their usefulness in applications requiring precise localization or understanding of object boundaries.

Some research has explored using CNN-based models for image segmentation, but many of these approaches focus on binary or coarse segmentation and are trained on limited or highly controlled datasets. Furthermore, traditional methods struggle to capture contextual relationships between different parts of an image, leading to reduced accuracy in complex scenes.

Disadvantages

- Limited pixel-level understanding
- Poor boundary localization
- Ineffective in complex or cluttered scenes
- Lack of contextual awareness
- High dependency on manually labeled datasets

2.2 Proposed System:

The proposed system addresses the limitations of traditional image analysis by using **semantic segmentation** to interpret images at the pixel level. The project implements deep learning-based segmentation architectures such as U-Net and DeepLabV3+, trained on datasets like PASCAL VOC and Cityscapes for multi-class segmentation tasks.

These models employ advanced techniques such as encoder-decoder structures, dilated (atrous) convolutions, and skip connections to preserve spatial resolution while capturing both local and global features. This allows for accurate segmentation of complex scenes involving multiple overlapping objects.

The system is designed to work on both images and video frames, enabling real-time applications. By providing a detailed semantic map of each frame, it supports applications in smart surveillance, autonomous navigation, and medical imaging.

Advantages

- Time Efficient.
- Cost Effective.
- Scalability.

Hardware and Software Requirements

Hardware Requirements

- Operating System : Windows 7 or7+
- processor : Pentium –IV
- RAM : 4GB (min)
- Hard Disk : 20GB
- Key Board : Standard Windows Keyboard
- Mouse : Two or Three Button Mouse
- Monitor : SVGA

Software Requirements

- Operating System : Windows 7 Ultimate
- Coding Language : Python.
- Front-End : Python.
- Back-End : Django-ORM
- Designing : Html, css, javascript.
- Data Base : MySQL (WAMP Server).

2.3. Feasibility Study

An important outcome of the preliminary investigation is the determination that the proposed system is feasible. A system is considered feasible only if it can be developed and deployed effectively within limited resources and time. The key types of feasibility considered in this project are:

- Operational Feasibility
- Economic Feasibility
- Technical Feasibility

Operational Feasibility

Operational feasibility focuses on how effectively the system meets its intended objectives in a real-world setting. The proposed semantic segmentation system helps automate complex image interpretation tasks that would otherwise require expert analysis. It significantly reduces the manual effort needed for labeling and analysis, making it especially useful for smart surveillance, medical diagnostics, and autonomous systems. Based on testing and usability assessment, the system is proven to be operationally feasible and practical for various use cases.

Economic Feasibility

Economic feasibility examines the cost-effectiveness of the proposed solution. The system is built using **open-source tools** like TensorFlow, PyTorch, and publicly available datasets such as **PASCAL VOC** and **Cityscapes**, minimizing software and licensing costs. As the models can be trained or fine-tuned on standard GPU-enabled systems or via cloud platforms, there is no need for highly specialized hardware. This ensures the project is economically viable and well-suited for institutional or academic use without incurring major expenses.

Technical Feasibility

Technical feasibility assesses whether the existing technology infrastructure and tools are sufficient to support the system. The solution uses well-supported deep learning frameworks and runs on commonly available hardware such as **GPU-enabled systems** or cloud-based platforms. It is implemented using **Python**, with supporting libraries like **Torchvision**, **OpenCV**, and **Matplotlib**. The system is platform-independent and can be integrated into web-based or desktop applications. Therefore, the technical requirements are well within reach, confirming the system's technical feasibility.

CHAPTER-III

SYSTEM DESIGN

3.1. Module Description:

The system is divided into several key modules to ensure scalability, maintainability, and ease of understanding. Each module is responsible for a specific function within the semantic segmentation pipeline.

User Module

- Handles user registration, login, and authentication.
- Manages role-based access for admin and general users.
- Provides a dashboard to track segmentation results and upload new images or video frames for analysis.

Model Training Module

- Enables the admin to train and evaluate segmentation models (e.g., U-Net, DeepLabV3+).
- Allows dataset selection (e.g., PASCAL VOC, Cityscapes) and configuration of hyperparameters.
- Displays training progress, loss graphs, and accuracy metrics (such as mIoU and pixel accuracy).
- Supports loading pre-trained weights for transfer learning and fine-tuning.

Semantic Segmentation Module

- Performs pixel-wise segmentation of images and video frames using deep learning models.
- Outputs segmented images with class-wise color labels (e.g., road, vehicle, person, sky).
- Integrates post-processing techniques (like thresholding or contour smoothing) for visual clarity.
- Supports both batch mode and real-time analysis.

Service Provider Module

- The service provider logs in using valid credentials.
- After successful login, the provider can:
  - Upload and manage custom datasets

• Initiate training/testing of models

• Monitor system usage statistics

• Manage user roles and access privileges

Remote User Access Module

- Provides external users with access to segmentation outputs.
- Enables uploading of new images or videos to be segmented remotely.
- Supports downloading processed results and visualizations.
- Useful for clients in domains like healthcare, traffic monitoring, and agriculture.

ER Diagram

The Entity–Relationship (ER) model is used to visually represent the **logical structure** of a database. It helps in designing the database before actual implementation by mapping entities, their attributes, and the relationships between them.

In the context of this project — *Unlocking Detailed Image Interpretation through Semantic Segmentation using AI* — the ER diagram helps outline how different components like users, datasets, segmentation results, and logs are connected within the database system.

Key Components of ER Model:

- **Entity Set:** Represents real-world objects (e.g., *User*, *Dataset*, *Model*, *Results*). Each entity corresponds to a table in the database.
- **Attributes:** Represent properties or characteristics of an entity (e.g., *username*, *email*, *accuracy_score*).
- **Relationship Set:** Shows associations between entities (e.g., a *User* uploads a *Dataset* or views *Results*).

This model serves as a blueprint for database implementation, ensuring data consistency and integrity.

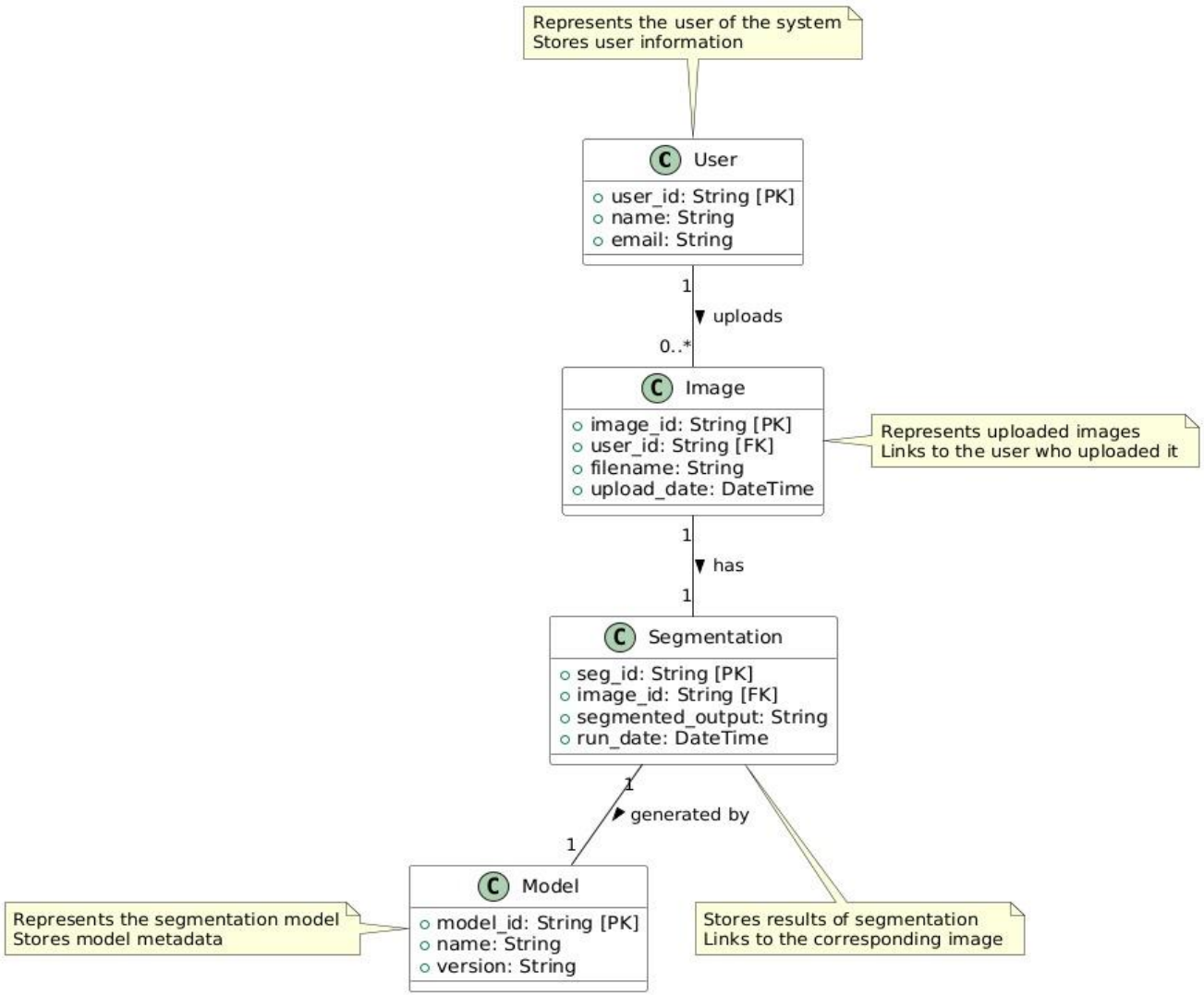


Figure3.2.1:ER Diagram

UML Diagram

UML stands for Unified Modeling Language. It is a standardized general-purpose modeling language used primarily in the field of object-oriented software engineering. UML helps developers, designers, and stakeholders to visually understand, design, and communicate the structure and behavior of a system.

UML serves as a common platform for designing software architecture by providing visual representations of system components, their interactions, and workflows. It enhances understanding

and simplifies the development of large and complex systems by breaking them down into diagrams that capture both static and dynamic aspects.

Key Features of UML:

- Specifies, visualizes, constructs, and documents the artifacts of a software system.
- Promotes best engineering practices for object-oriented analysis and design.
- Uses graphical notations to represent components, relationships, and processes.
- Supports both software and non-software systems, including business process modeling.

Core UML Diagram Types (used in this project):

1. **Use Case Diagram**
    - Describes the interactions between users (actors) and the system.
    - Useful for representing system functionality from an external viewpoint.
2. **Class Diagram**
    - Shows the static structure of the system, including classes, attributes, and relationships.
    - Ideal for modeling the internal logic and database structure.
3. **Sequence Diagram**
    - Represents the flow of logic and communication between objects over time.
    - Helps understand how segmentation requests are processed.

Goals:

The primary design goals of the Unified Modeling Language (UML) are aimed at making it a powerful, flexible, and universally accepted modeling language for object-oriented systems. The goals are as follows:

- Provide an expressive visual modeling language
  To enable users to create meaningful, clear, and standardized models that represent the structure and behavior of software systems.
- Support extendibility and specialization
  UML allows for the extension of core concepts through mechanisms like stereotypes, tagged values, and profiles to adapt to specific domains.
- Remain independent of programming languages and processes
  UML is designed to be used across multiple platforms and does not depend on any one programming language or development methodology.
- Offer a formal foundation
  UML provides a well-defined semantic and syntactic basis to ensure consistent interpretation of models.
- Encourage object-oriented tool development
  The adoption of UML fosters the growth of tools that support model-driven development, increasing efficiency and automation.
- Support higher-level development concepts
  UML facilitates the use of advanced software engineering principles like frameworks, design patterns, collaborations, and component-based development.
- Integrate industry best practices
  UML is a culmination of successful practices from leading methodologies (like Booch, OMT, and OOSE), making it a reliable and robust modeling standard.

Use Case Diagram:

A Use Case Diagram in the Unified Modeling Language (UML) is a type of behavioral diagram that provides a graphical overview of the interactions between users (actors) and the system. It highlights the functionalities (use cases) that the system offers and how external entities interact with them.

The primary goal of a use case diagram is to describe:

- What functions the system performs
- Which users or actors initiate these functions
- The relationships or dependencies between these functions

In this project, the use case diagram helps visualize the interactions between users, admin, and the system that performs semantic segmentation.

Main Actors:

- **Admin:** Responsible for training/testing models, managing datasets, and viewing analytics.
- **User:** Uploads images or videos and views segmentation results.
- **Remote Client:** Accesses processed outputs and predictions from external locations.

Primary Use Cases:

- Register/Login
- Upload Images or Video Frames
- Train Segmentation Model
- Test Model on Dataset
- View Accuracy Graphs
- Download Segmentation Results
- Manage Datasets

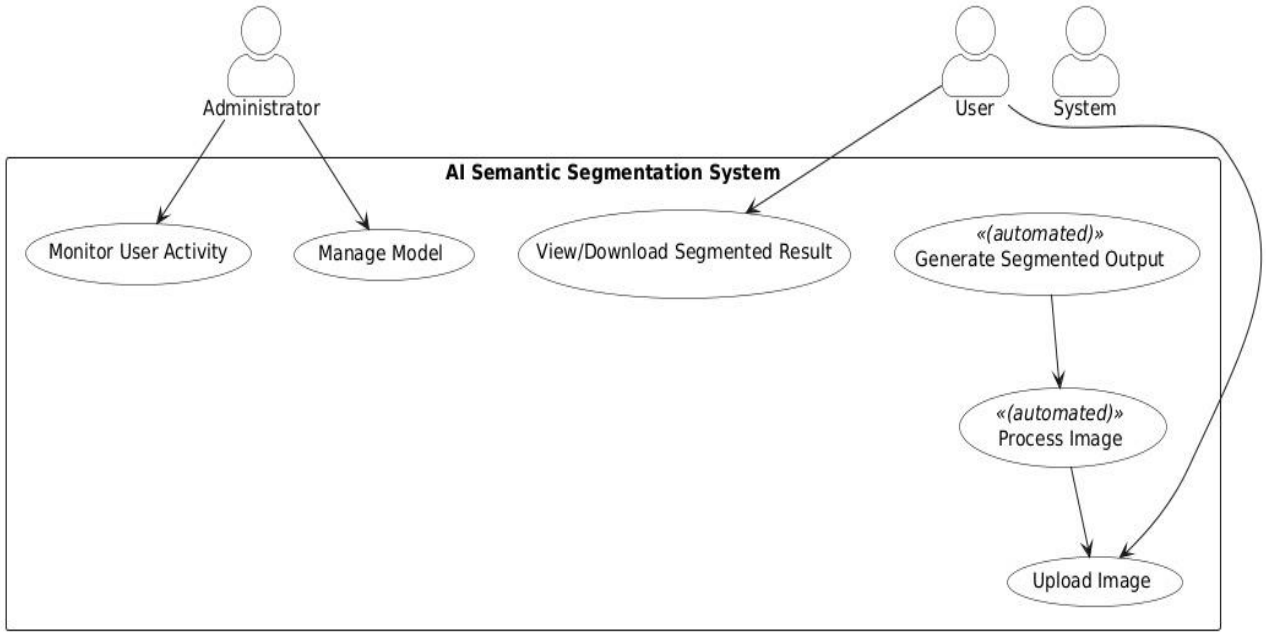


Fig 3.3.1: Use Case Diagram

A Use Case Diagram is a high-level representation of the functional behavior of a system. It shows what the system does from the user's perspective, who the users are (actors), and how they interact with the system. This diagram does not describe the internal logic but focuses on external interactions and the scope of the system functionalities.

In this project, the system is designed for semantic segmentation of images and video frames using AI. The main actors are the System and the User, each with specific roles and interactions.

Class Diagram

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

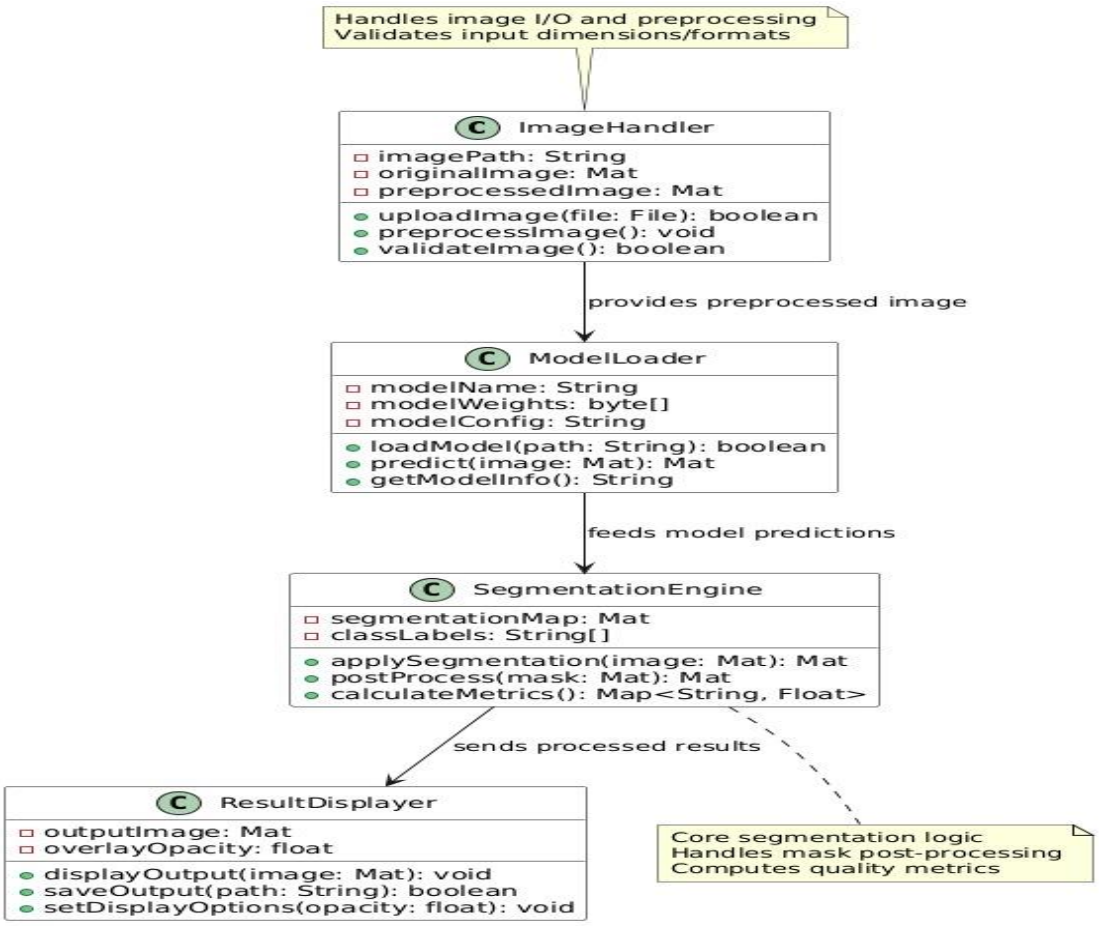


Fig.3.3.2. Class Diagram

A class diagram is an illustration of the relationships and source code dependencies among classes in the Unified Modeling Language (UML). In this context, a class defines the methods and variables in an object, which is a specific entity in a program or the unit of code representing that entity.

In the class diagram we have two classes user and system and the user functions are choose the data, check the preprocessed data, choose the algorithm, apply the trained model on data and test and classify the data finally view results. The system functions are upload the dataset and preprocess the dataset, split the data for train and test, train the data, train the model and predict the results.

Sequence Diagram

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagram.

The sequence diagram has also two classes user and system here the user will choose the dataset and upload on the system. After uploading the dataset user will check the preprocessed dataset and the system will preprocess the data, user will choose the algorithm through the system and the system will split the data for train and test. After system train data the user will apply the trained model on data through system. The system will train the data and model, user will test and classify the data and finally predict the results in sequence diagram and finally user will be viewed the results.

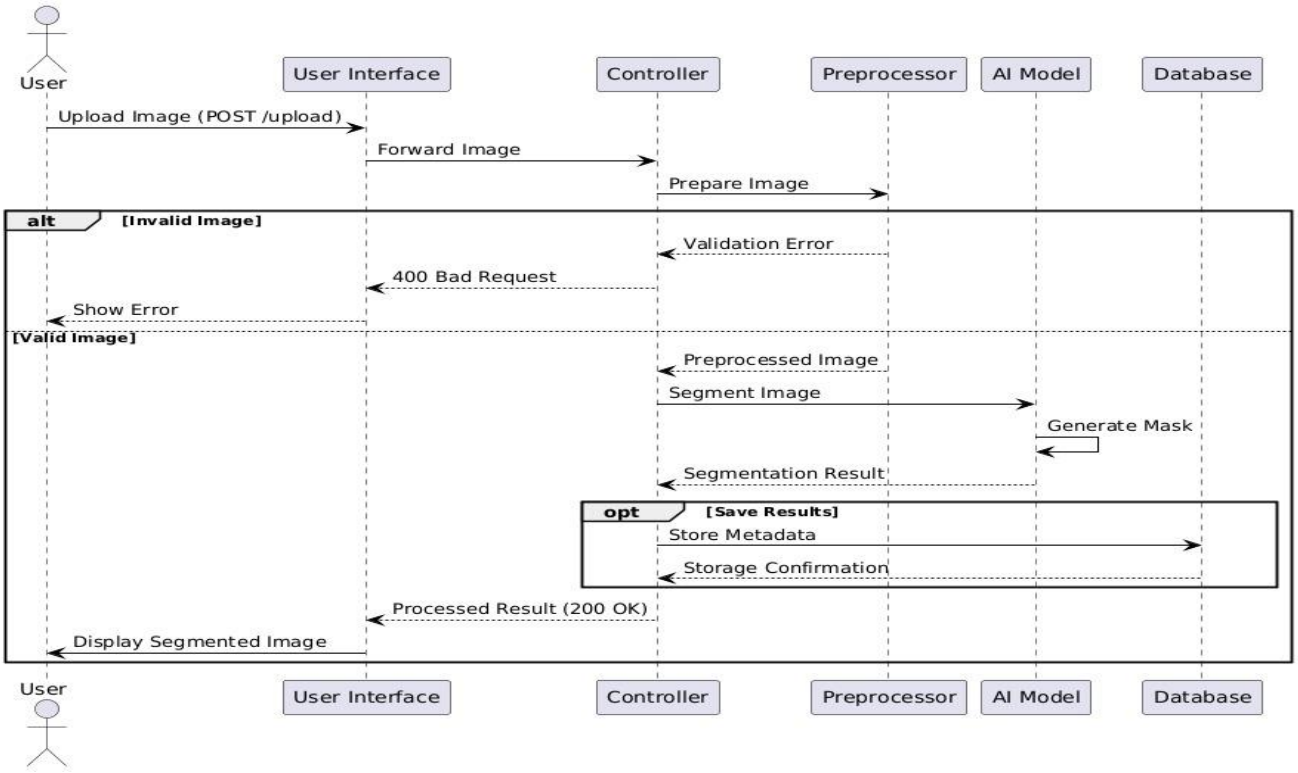


Fig.3.3.2. Sequence Diagram

Activity Diagram:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of

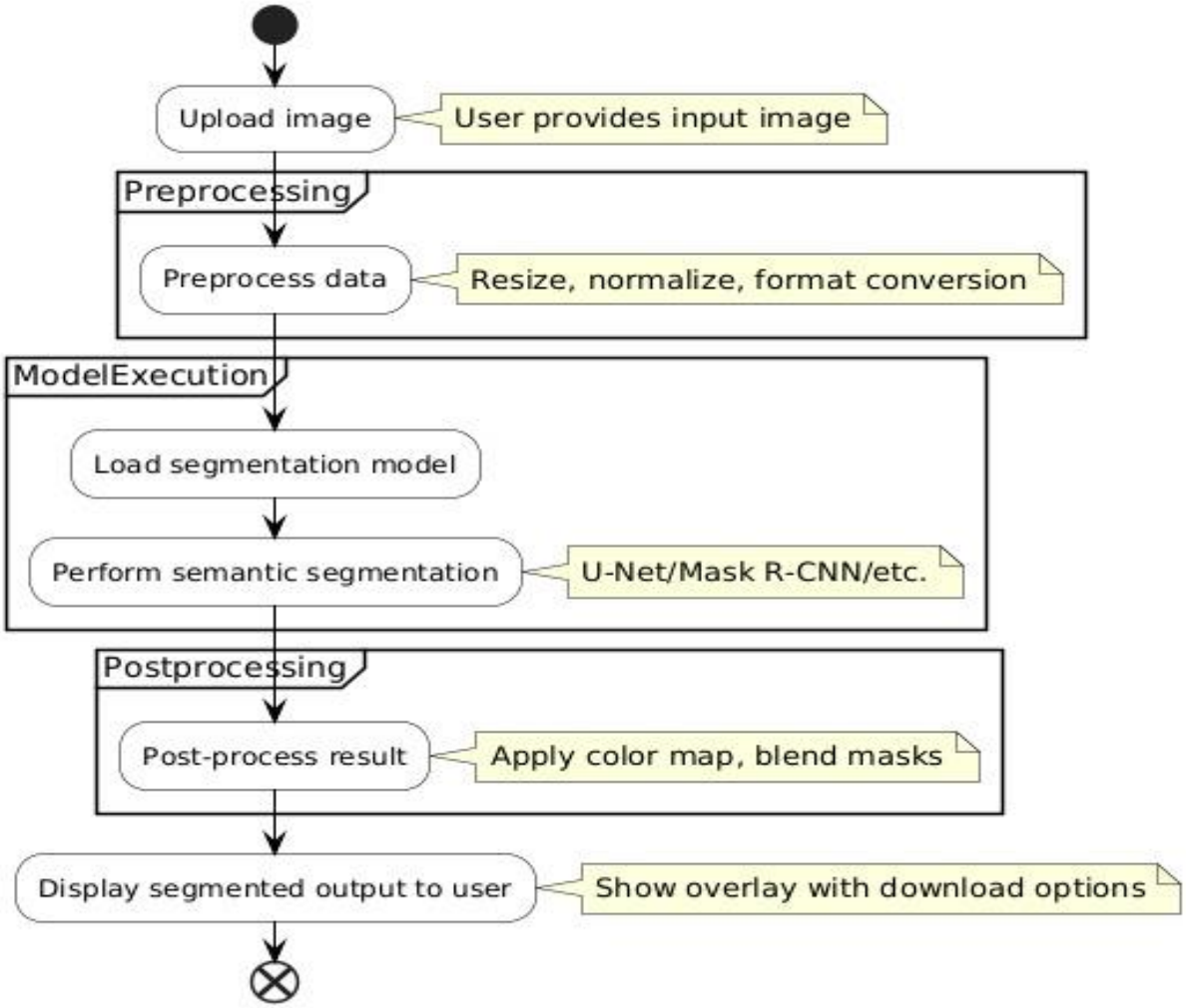


Fig 3.3.4: Activity Diagram

Activity diagrams are similar to flowcharts because they show the flow between the actions in an activity; however, activity diagrams can also show parallel or concurrent flows and alternate flows. In activity diagrams, you use activity nodes and activity edges to model the

In activity diagram the user and system will follows the step by step process. User will choose

the dataset and check the preprocessed dataset and choose the algorithm. The user will apply the trained model on data and test and classify the data. The system flows to upload the dataset and preprocess the dataset and split the data for train and test, the train the data and train the model has to predict the results finally in activity diagram.

Component Diagram:

A component diagram, also known as a UML component diagram, describes the organization and wiring of the physical components in a system. Component diagrams are often drawn to help model implementation details and double-check that every aspect of the system's required functions is covered by planned development.

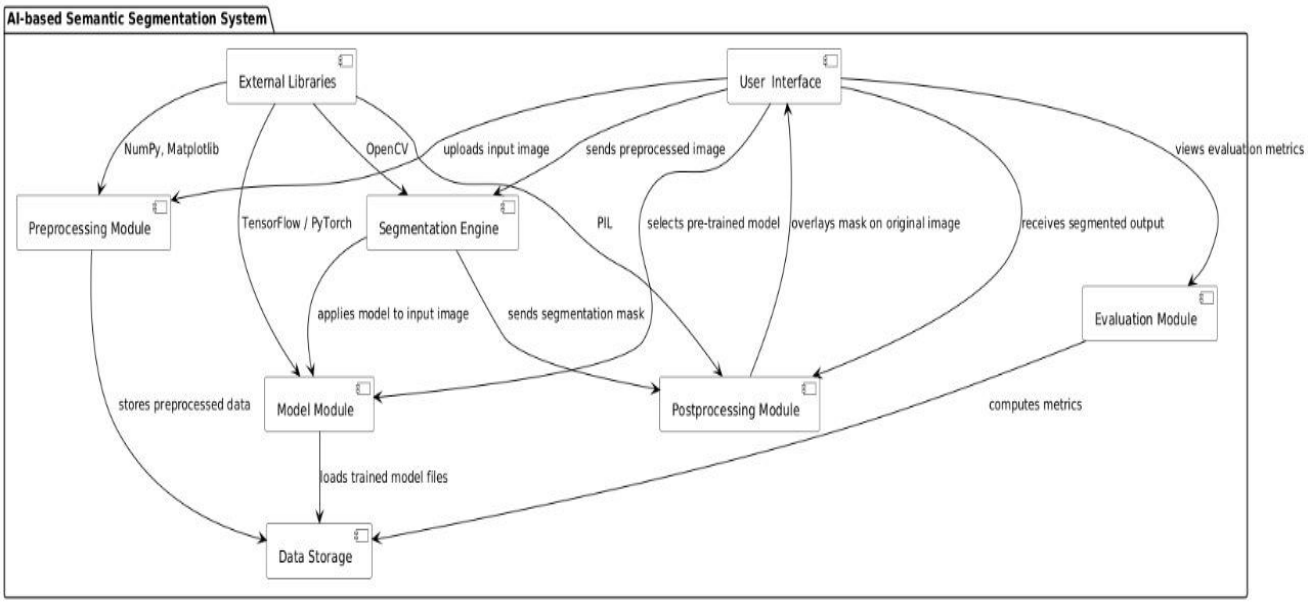


Fig 3.3.5: Component Diagram

Collaboration Diagram:

In collaboration diagram the method call sequence is indicated by some technique as shown below. The number indicates how the methods are called one after another. We have taken the same order management system to describe the collaboration diagram. The method calls are similar to that of a sequence diagram. But the difference is that the sequence diagram

does not describe the object organization where as the collaboration diagram shows the object organization.

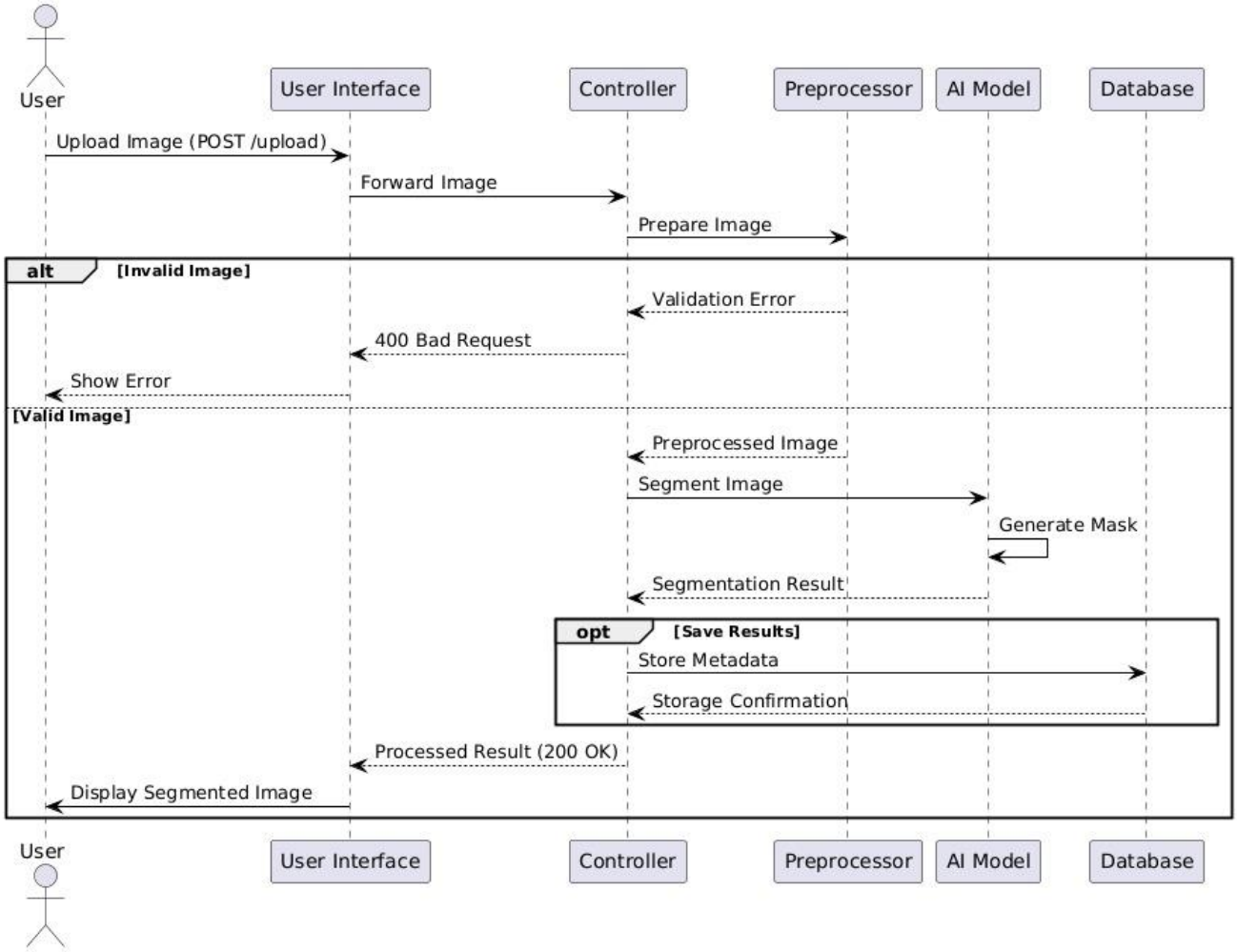


Fig 3.3.6: Collaboration Diagram

A collaboration diagram, also known as a communication diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). These diagrams can be used to portray the dynamic behavior of a particular use case and define the role of each object.

Deployment Diagram:

Deployment diagram represents the deployment view of a system. It is related to the component diagram. Because the components are deployed using the deployment diagrams. A deployment diagram consists of nodes. Nodes are nothing but physical hard wares used to deploy the application.

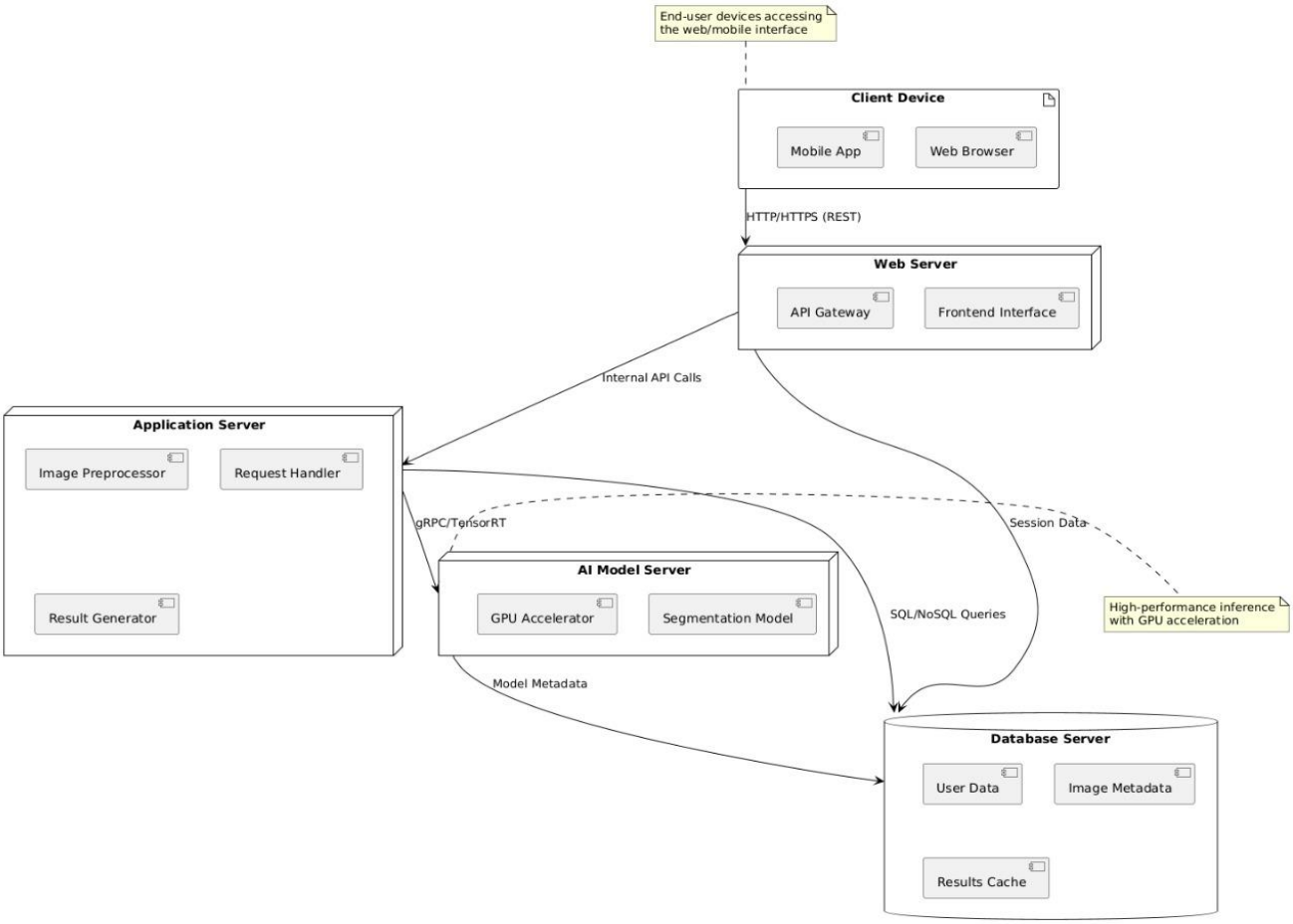


Fig 3.3.7: Deployment Diagram

CHAPTER-IV

SYSTEM IMPLEMENTATION

4.1 Language Selection:

4.1. INTRODUCTION TO PYTHON

What Is a Script?

You can use Microsoft Notepad, Microsoft WordPad, Microsoft Word, or just about any word processor if you want to creating Python script files.

Difference between a script and a program Script?

Up to this point, I have concentrated on the interactive programming capability of Python. This is a very useful capability that allows you to type in a program and to have it executed immediately in an interactive mode

Scripts are reusable

Basically, a script is a text file containing the statements that comprise a Python program. Once you have created the script, you can execute it over and over without having to retype it each time.

Scripts are editable

Perhaps, more importantly, you can make different versions of the script by modifying the statements from one file to the next using a text editor. Then you can execute each of the individual versions. In this way, it is easy to create different programs with a minimum amount of typing.

You will need a text editor

Just about any text editor will suffice for

Scripts are distinct from the core code of the application, which is usually written in a different language, and are often created or at least modified by the end-user. Scripts are often interpreted from source code or byte code, where as the applications the control are traditionally compiled to native machine code.

Program:

The program has an executable form that the computer can use directly to execute the instructions. The same program in its human-readable source code form, from which executable programs are derived (e.g., compiled)

Python

what is Python? Chances you are asking yourself this. You may have found this book because you want to learn to program but don't know anything about programming languages. Or you may have heard of programming languages like C, C++, C#, or Java and want to know what Python is and how it compares to "big name" languages. Hopefully I can explain it for you.

Python concepts

If you're not interested in the the how's and whys of Python, feel free to skip to the next chapter. In this chapter I will try to explain to the reader why I think Python is one of the best languages available and why it's a great one to start programming with.

Open-source general-purpose language.

Object Oriented, Procedural, Functional

Easy to interface with C/ObjC/Java/Fortran

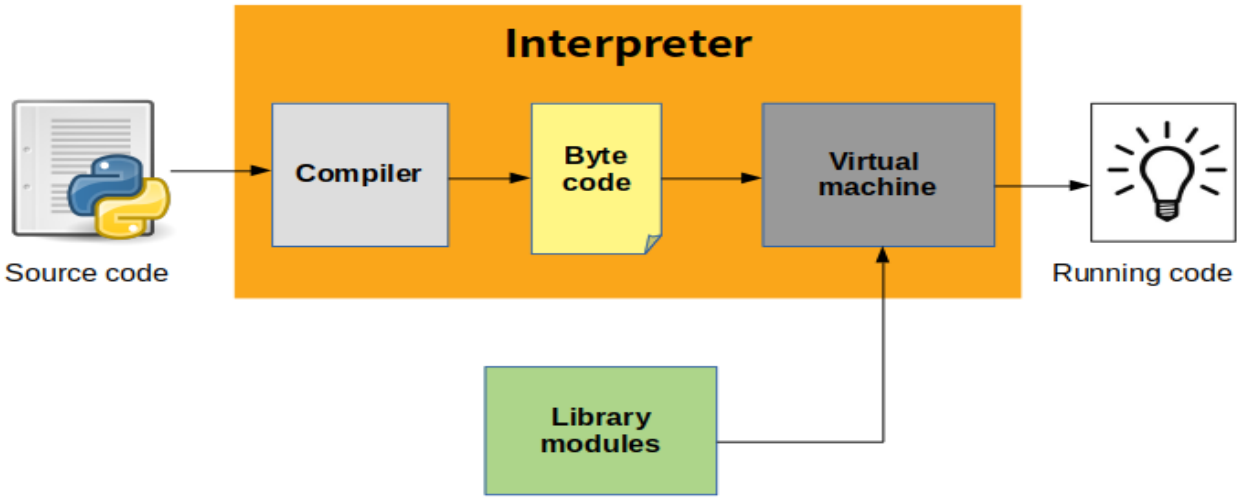
Easy-ish to interface with C++ (via SWIG)

Great interactive environment

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

Python is Interpreted − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.

to WWW browsers to games.



Standard Data Types

The data stored in memory can be of many types. For example, a person's age is stored as a numeric value and his or her address is stored as alphanumeric characters. Python has various standard data types that are used to define the operations possible on them and the storage method for each of them.

Python has five standard data types −

Numbers

String

List

Tuple

Dictionary

Python Numbers

Number data types store numeric values. Number objects are created when you assign a value to them.

Python Strings

Strings in Python are identified as a contiguous set of characters represented in the quotation marks. Python allows for either pairs of single or double quotes. Subsets of strings can be taken using the slice operator ([ ] and [:]) with indexes starting at 0 in the beginning of the string and working their way from -1 at the end.

Python Lists

Lists are the most versatile of Python's compound data types. A list contains items separated by commas and enclosed within square brackets ([]). To some extent, lists are similar to arrays in C. One difference between them is that all the items belonging to a list can be of different data type.

The values stored in a list can be accessed using the slice operator ([ ] and [:]) with indexes starting at 0 in the beginning of the list and working their way to end -1. The plus (+) sign is the list concatenation operator, and the asterisk (*) is the repetition operator.

Python Tuples

A tuple is another sequence data type that is similar to the list. A tuple consists of a number of values separated by commas. Unlike lists, however, tuples are enclosed within parentheses.

The main differences between lists and tuples are: Lists are enclosed in brackets ([ ]) and their elements and size can be changed, while tuples are enclosed in parentheses (( )) and cannot be updated. Tuples

can be thought of as read-only lists.

Python Dictionary

Python's dictionaries are kind of hash table type. They work like associative arrays or hashes found in Perl and consist of key-value pairs. A dictionary key can be almost any Python type, but are usually numbers or strings. Values, on the other hand, can be any arbitrary Python object.

Dictionaries are enclosed by curly braces ({ }) and values can be assigned and accessed using square braces ([]).

Different modes in python

Python has two basic modes: normal and interactive.

The normal mode is the mode where the scripted and finished .py files are run in the Python interpreter. Interactive mode is a command line shell which gives immediate feedback for each statement, while running previously fed statements in active memory. As new lines are fed into the interpreter, the fed program is evaluated both in part and in whole

20 Python libraries

Requests. The most famous http library written by Kenneth Reitz. It’s a must have for every python developer.

Numpy

NumPy’s main object is the homogeneous multidimensional array. It is a table of elements (usually numbers), all of the same type, indexed by a tuple of positive integers. In NumPy dimensions are called axes. The number of axes is rank.

• Offers Matlab-ish capabilities within Python

• Fast array operations

• 2D arrays, multi-D arrays, linear algebra etc.

matplotlib

• High quality plotting library.

Python class and objects

Python modules

Python allows us to store our code in files (also called modules). This is very useful for more serious programming, where we do not want to retype a long function definition from the very beginning just to change one mistake. In doing this, we are essentially defining our own modules, just like the modules defined already in the Python library.

To support this, Python has a way to put definitions in a file and use them in a script or in an interactive instance of the interpreter. Such a file is called a module; definitions from a module can be imported into other modules or into the main module.

MySQL Connector/Python includes support for:

Almost all features provided by MySQL Server up to and including MySQL Server version 5.7.

Converting parameter values back and forth between Python and MySQL data types, for example Python datetime and MySQL DATETIME. You can turn automatic conversion on for convenience, or off for optimal performance.

All MySQL extensions to standard SQL syntax.

Protocol compression, which enables compressing the data stream between the client and server.

Connections using TCP/IP sockets and on Unix using Unix sockets.

Secure TCP/IP connections using SSL.

Self-contained driver. Connector/Python does not require the MySQL client library or any Python modules outside the standard library.

MYSQL:

MySQL is a Relational Database Management System (RDBMS) whereas the structured Query Language (SQL) is the language used for handling the RDBMS using commands i.e Creating, Inserting, Updating and Deleting the data from the databases. SQL commands are case insensitive i.e CREATE and create signify the same command.

In this article, we will be discussing the MySQL Connector module of Python, how to install this module and a piece of code on how to connect this with the MySQL database. For any application, it is very important to store the database on a server for easy data access.

What is MYSQL Connector/Python?

MySQL Connector/Python enables Python programs to access MySQL databases, using an API that is compliant with the Python Database API Specification v2.0 (PEP 249). It is written in pure Python and does not have any dependencies except for the Python Standard Library.

4.2 SCREEN SHOTS

All the screen shots of the project are given below which shows the execution for better understanding of the project.

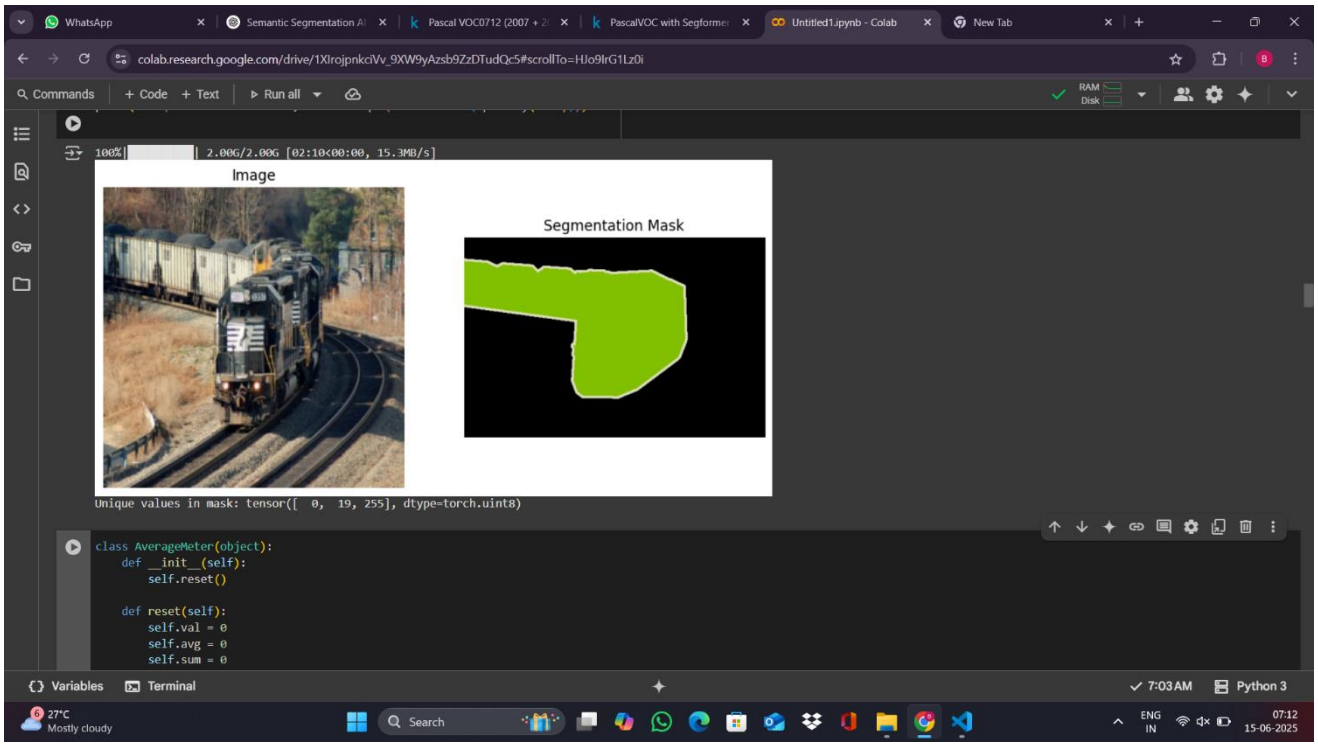


Figure 4.2.1 Unique values in mask

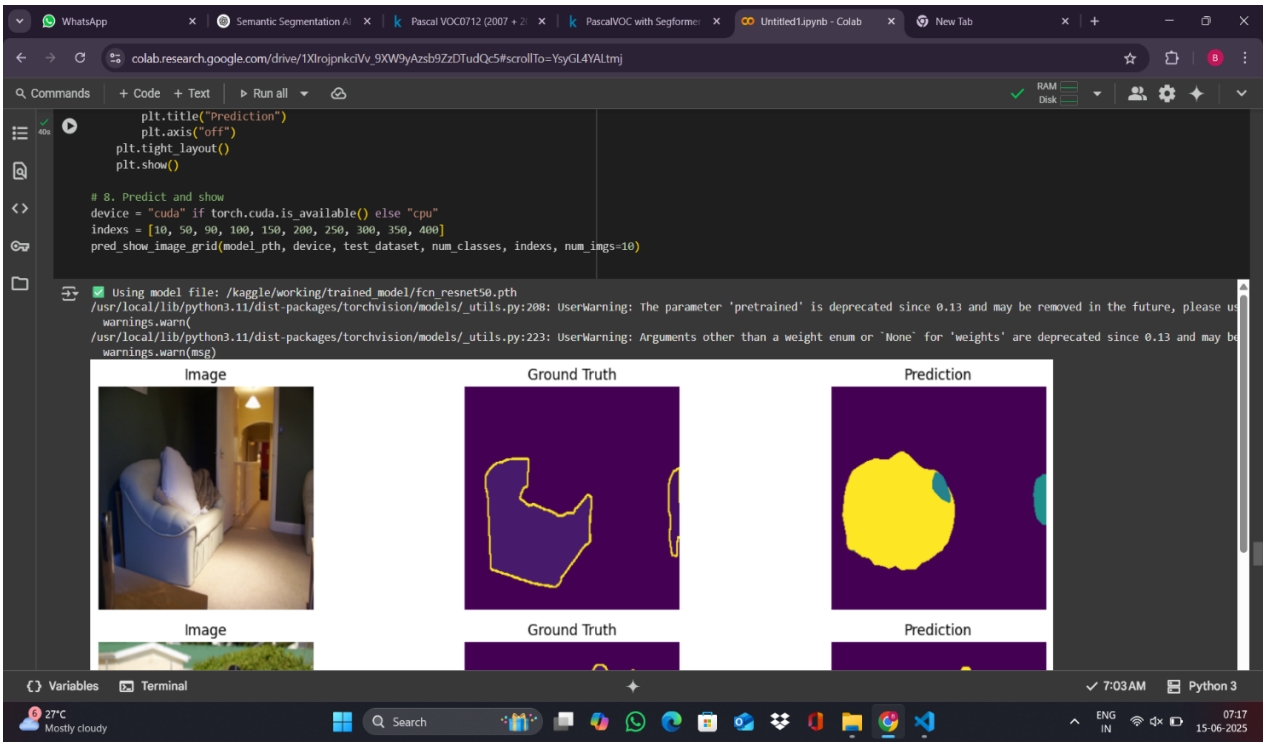


Figure 4.2.1.1 Interpretation results

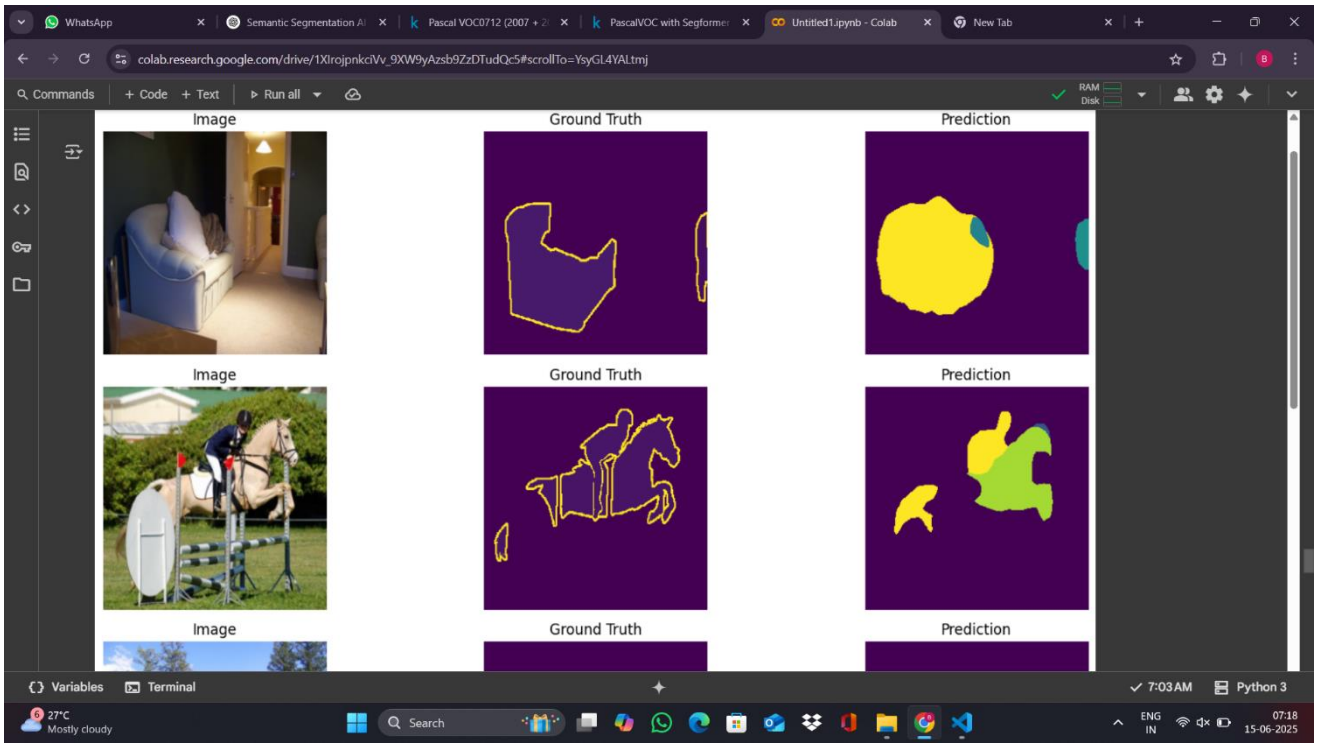


Figure 4.2.2 Interpretation results

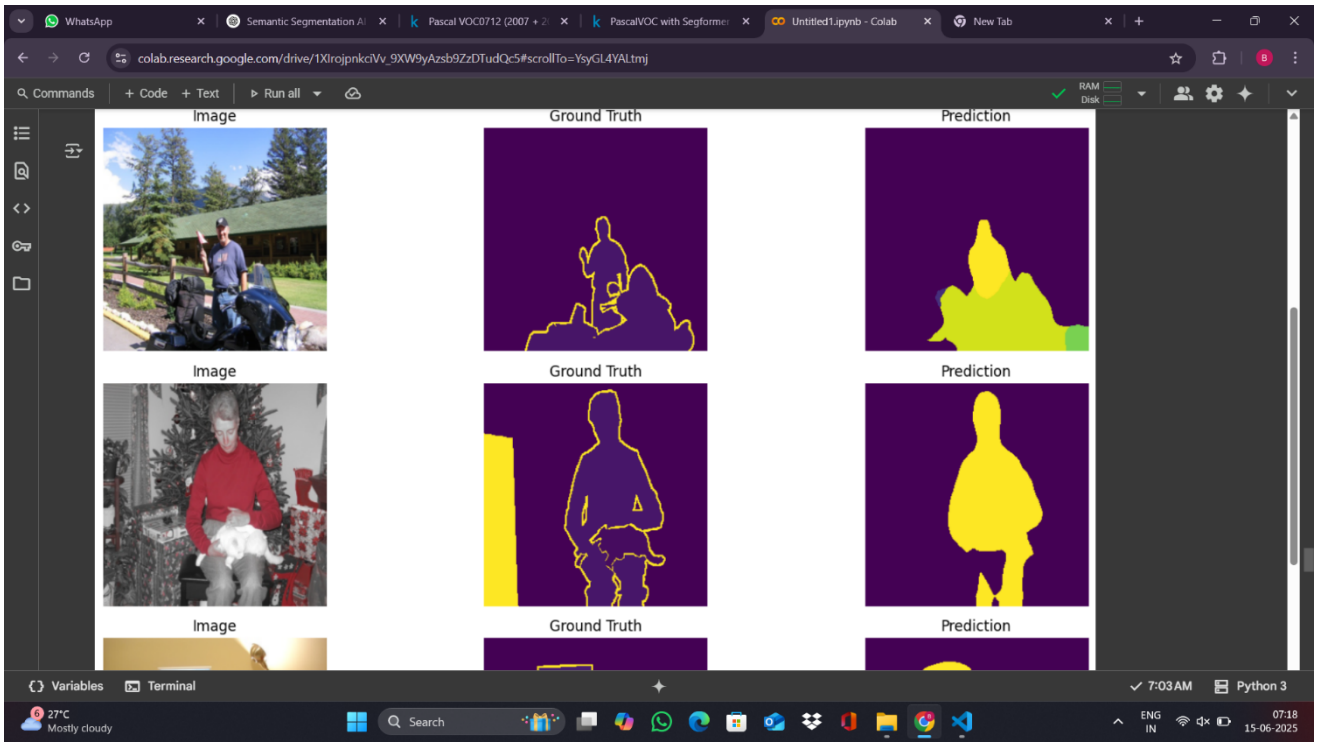


Figure 4.2.2.1 Interpretation results

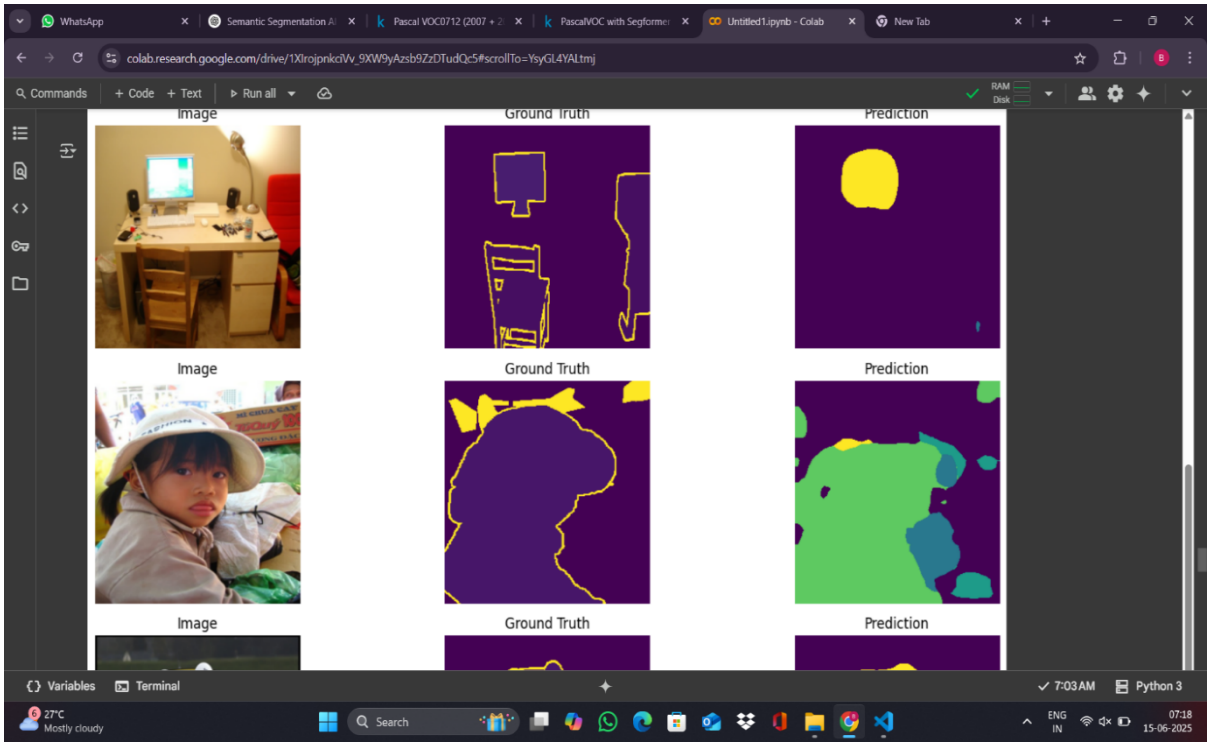


Figure 4.2.3 Segmentation results

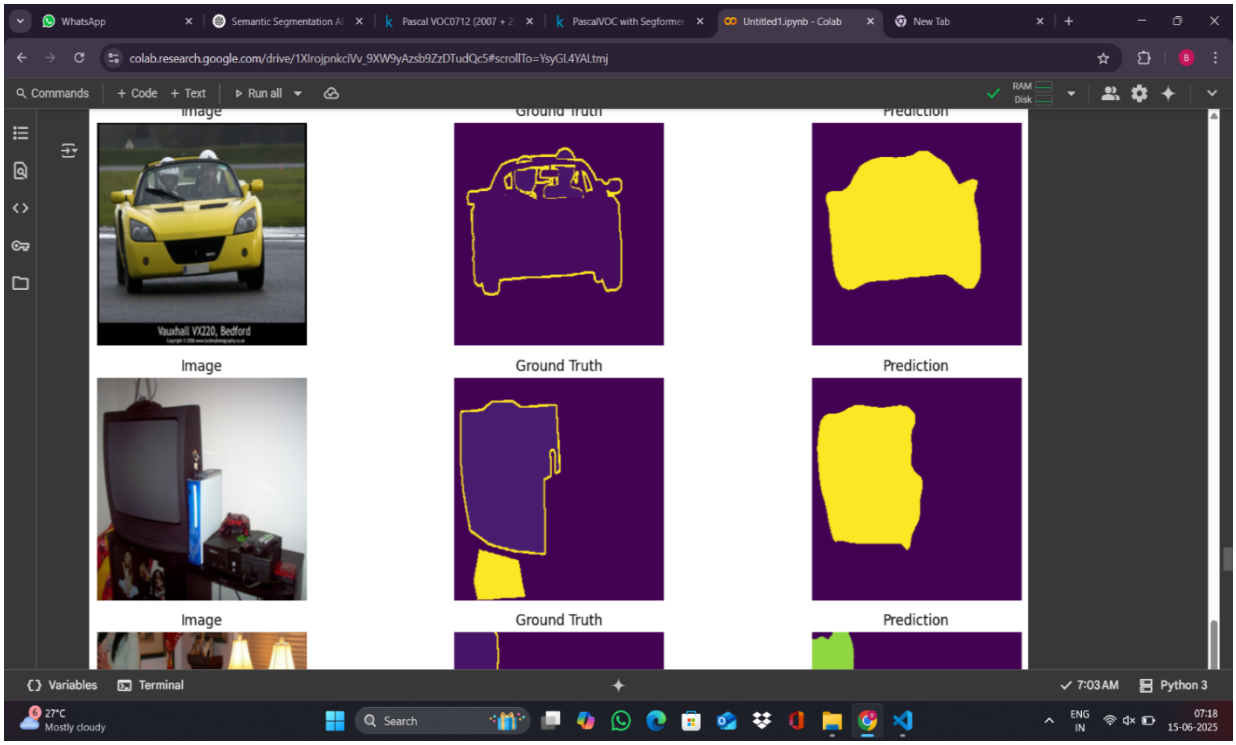


Figure 4.2.4 Segmentation results

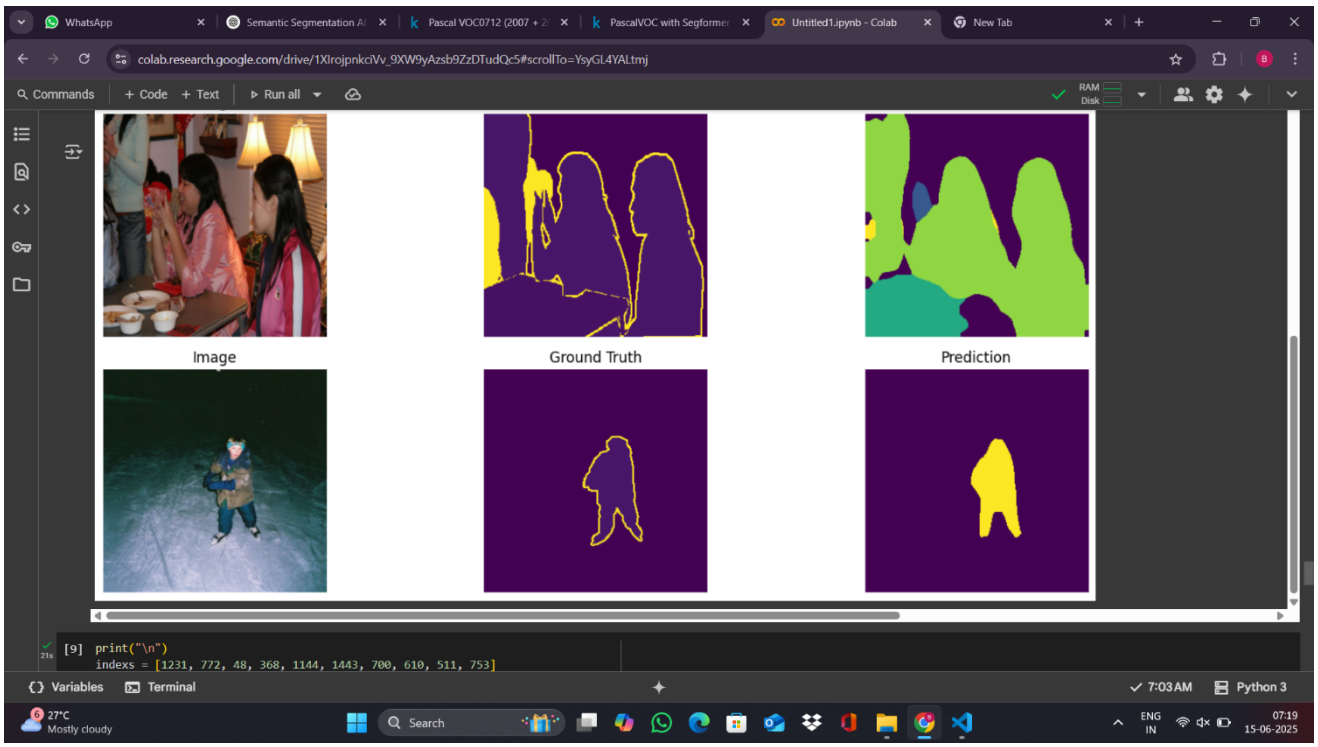


Figure 4.2.5 Segmentation results

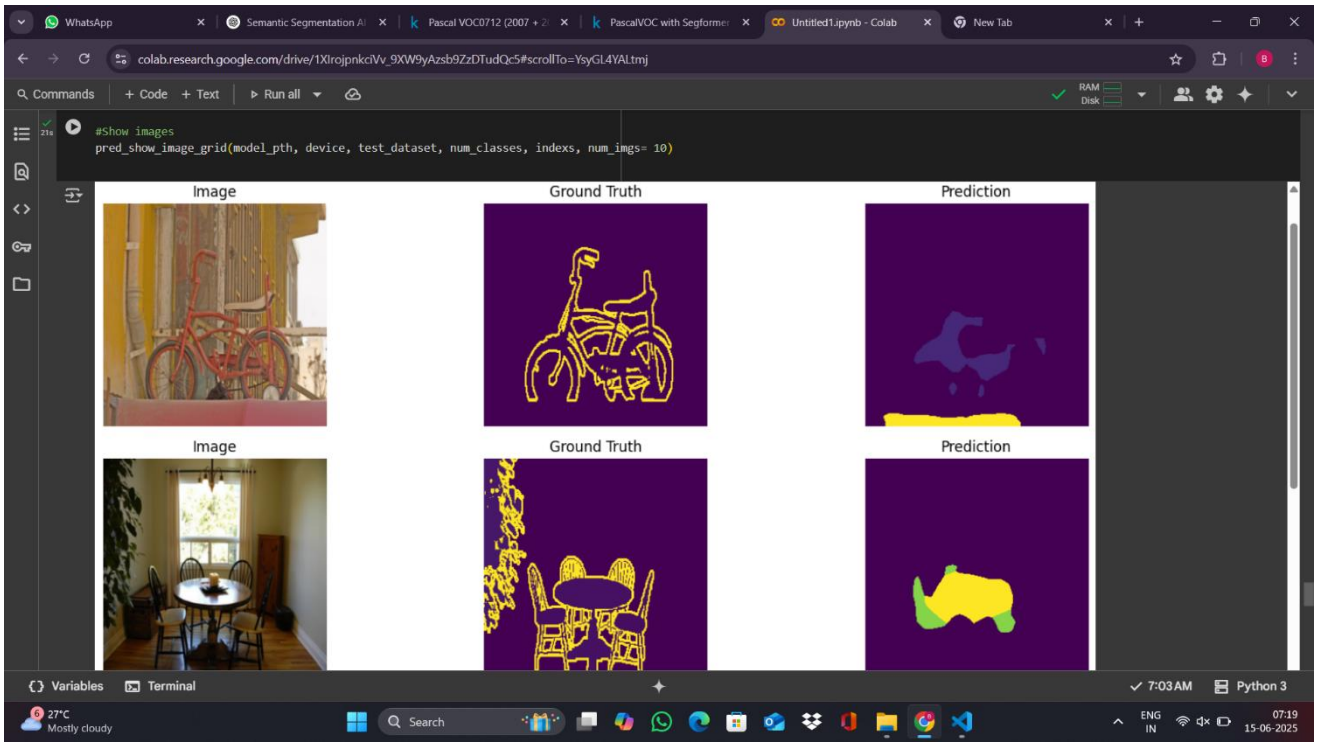


Figure 4.2.6 Segmentation results

4.3. SAMPLE CODE

Sample code has been given to show how the system has been implemented. The methodology used to create the screens and the sample JavaScript code is given here.

```
!pip install transformers
!pip install segmentation_models_pytorch
!pip install torchmetrics
import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from torch.utils.tensorboard import SummaryWriter

# Use only JaccardIndex (IOU)
from torchmetrics.classification import JaccardIndex

import segmentation_models_pytorch as smp
import albumentations as A
from albumentations.pytorch import ToTensorV2
from transformers import SegformerForSemanticSegmentation

from torchvision.datasets import VOCSegmentation
from tqdm import tqdm
import os
import numpy as np
import matplotlib.pyplot as plt
import cv2
import warnings
import shutil

warnings.filterwarnings("ignore")

# Manual Dice Score function (works for binary and multiclass if adapted)
def dice_score(preds, targets, smooth=1e-6):
    preds = preds.view(-1)
    targets = targets.view(-1)
    intersection = (preds * targets).sum()
    dice = (2. * intersection + smooth) / (preds.sum() + targets.sum() + smooth)
    return dice
cv2.setNumThreads(0)
cv2.ocl.setUseOpenCL(False)

VOC_CLASSES = [
    "background",
    "aeroplane",
    "bicycle",
    "bird",
```

```
    "boat",
    "bottle",
    "bus",
    "car",
    "cat",
    "chair",
    "cow",
    "diningtable",
    "dog",
    "horse",
    "motorbike",
    "person",
    "potted plant",
    "sheep",
    "sofa",
    "train",
    "tv/monitor",
]

VOC_COLORMAP = [
    [0, 0, 0],
    [128, 0, 0],
    [0, 128, 0],
    [128, 128, 0],
    [0, 0, 128],
    [128, 0, 128],
    [0, 128, 128],
    [128, 128, 128],
    [64, 0, 0],
    [192, 0, 0],
    [64, 128, 0],
    [192, 128, 0],
    [64, 0, 128],
    [192, 0, 128],
    [64, 128, 128],
    [192, 128, 128],
    [0, 64, 0],
    [128, 64, 0],
    [0, 192, 0],
    [128, 192, 0],
    [0, 64, 128],
]

class PascalVOCSearchDataset(VOCSegmentation):
    def __init__(self, root="/kaggle/input/voc0712/VOC_dataset", year="2012", image_set="train",
download=False, transform=None):
        super().__init__(root=root, year=year, image_set=image_set, download=download,
```

```
transform=transform)

    @staticmethod
    def _convert_to_segmentation_mask(mask):
        # This function converts a mask from the Pascal VOC format to the format required by
AutoAlbument.
        #
        # Pascal VOC uses an RGB image to encode the segmentation mask for that image. RGB values
of a pixel
        # encode the pixel's class.
        #
        # AutoAlbument requires a segmentation mask to be a NumPy array with the shape [height,
width, num_classes].
        # Each channel in this mask should encode values for a single class. Pixel in a mask channel
should have
        # a value of 1.0 if the pixel of the image belongs to this class and 0.0 otherwise.
        height, width = mask.shape[:2]
        segmentation_mask = np.zeros((height, width, len(VOC_COLORMAP)), dtype=np.float32)
        for label_index, label in enumerate(VOC_COLORMAP):
            segmentation_mask[:, :, label_index] = np.all(mask == label, axis=-1).astype(float)
        return segmentation_mask

    def __getitem__(self, index):
        image = cv2.imread(self.images[index])
        image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
        mask = cv2.imread(self.masks[index])
        mask = cv2.cvtColor(mask, cv2.COLOR_BGR2RGB)
        mask = self._convert_to_segmentation_mask(mask)
        if self.transform is not None:
            transformed = self.transform(image=image, mask=mask)
            image = transformed["image"]
            mask = transformed["mask"]
        return image.float(), mask.argmax(dim=2).squeeze().long()
class UnNormalize(object):
    def __init__(self, mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]):
        self. mean = mean
        self.std = std

    def __call__(self, tensor):
        for t, m, s in zip(tensor, self.mean, self.std):
            t.mul_(s).add_(m)

        return tensor

unorm = UnNormalize()
size = 256
train_transform = A.Compose([
```

```
    A.Resize(size, size),
    A.RandomRotate90(),
    A.GaussianBlur(),
    A.CLAHE(clip_limit=4.0, tile_grid_size=(8, 8), p=0.5),
    A.HueSaturationValue(hue_shift_limit=20, sat_shift_limit=30, val_shift_limit=20, p=0.5),
    A.HorizontalFlip(),
    A.CoarseDropout(max_height=25, max_width=25, fill_value=255),
    A.GridDistortion(num_steps=5, distort_limit=0.3),
    A.ElasticTransform(alpha=1.0),
    A.Perspective(scale=(0.05, 0.1)),
    A.Normalize(mean=(0.485, 0.456, 0.406), std=(0.229, 0.224, 0.225), max_pixel_value=255.0),
    ToTensorV2()
])

test_transform = A.Compose([
    A.Resize(size, size),
    A.Normalize(mean = (0.485, 0.456, 0.406), std = (0.229, 0.224, 0.225), max_value_pixel = 255.0),
    ToTensorV2()
])
from torchvision.datasets import VOCSegmentation
import torchvision.transforms as T

# Define your root directory
voc_root = "./data"  # You can change this path

# Optional image transformations
train_transform = T.Compose([
    T.ToTensor(),
    T.Resize((256, 256)),
])

# Load Pascal VOC 2012 dataset (will download if not found)
train_dataset = VOCSegmentation(
    root=voc_root,
    year='2012',
    image_set='train',
    download=True,
    transform=train_transform,
    target_transform=None
)

test_dataset = VOCSegmentation(
    root=voc_root,
    year='2012',
    image_set='val',
    download=True,
    transform=train_transform,
```

```
    target_transform=None
)

# Un-normalize function for visualization (if needed)
def unorm(img_tensor):
    return img_tensor * 255  # simple inverse if just ToTensor()

# Show a sample image and mask
image, mask = train_dataset[10]
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(unorm(image).permute(1, 2, 0).numpy().astype("uint8"))
plt.title("Image")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(mask)
plt.title("Segmentation Mask")
plt.axis("off")
plt.show()

print("Unique values in mask:", torch.unique(torch.tensor(np.array(mask))))
class AverageMeter(object):
    def __init__(self):
        self.reset()

    def reset(self):
        self.val = 0
        self.avg = 0
        self.sum = 0
        self.count = 0

    def update(self, val, n = 1):
        self.val += val
        self.sum += val * n
        self.count += n
        self.avg = self.sum / self.count
def plot_segmentation_metrics(train_loss, val_loss, train_mean_iou=None, val_mean_iou=None,
                              train_dice_score=None, val_dice_score=None):
    """
    Plot training and validation metrics (loss, mean IoU, Dice score) over epochs.

    Args:
    - train_loss (list): List of training loss values over epochs.
    - val_loss (list): List of validation loss values over epochs.
    - train_mean_iou (list, optional): List of training mean IoU values over epochs.
    - val_mean_iou (list, optional): List of validation mean IoU values over epochs.
    - train_dice_score (list, optional): List of training Dice score values over epochs.
```

```
    - val_dice_score (list, optional): List of validation Dice score values over epochs.
    """
    epochs = range(1, len(train_loss) + 1)

    plt.figure(figsize=(15, 10))

    # Plotting the Loss
    plt.subplot(2, 2, 1)
    plt.plot(epochs, train_loss, 'r', label='Training Loss')
    plt.plot(epochs, val_loss, 'b', label='Test Loss')
    plt.title('Training and Test Loss')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()

    # Plotting the Mean IoU (if available)
    if train_mean_iou and val_mean_iou:
        plt.subplot(2, 2, 2)
        plt.plot(epochs, train_mean_iou, 'r', label='Training Mean IoU')
        plt.plot(epochs, val_mean_iou, 'b', label='Test Mean IoU')
        plt.title('Training and Test Mean IoU')
        plt.xlabel('Epochs')
        plt.ylabel('Mean IoU')
        plt.legend()

    # Plotting the Dice Score (if available)
    if train_dice_score and val_dice_score:
        plt.subplot(2, 2, 3)
        plt.plot(epochs, train_dice_score, 'r', label='Training Dice Score')
        plt.plot(epochs, val_dice_score, 'b', label='Test Dice Score')
        plt.title('Training and Test Dice Score')
        plt.xlabel('Epochs')
        plt.ylabel('Dice Score')
        plt.legend()

    plt.tight_layout()
    plt.show()
upsample = nn.Upsample(scale_factor=4, mode='bilinear')
def train(num_class, batch_size, lr, epochs, checkpoint, trained_model):
    device = "cuda" if torch.cuda.is_available() else "cpu"
    n_workers = os.cpu_count()

    train_dataset = PascalVOCSearchDataset(transform=train_transform)
    train_loader = DataLoader(
        dataset=train_dataset,
        batch_size=batch_size,
        shuffle=True,
```

```
        num_workers=n_workers,
        drop_last=True
    )

    test_dataset = PascalVOCSearchDataset(image_set="val", transform=test_transform)
    test_loader = DataLoader(
        dataset=test_dataset,
        batch_size=batch_size,
        num_workers=n_workers,
        drop_last=True
    )

    model_name = "nvidia/segformer-b5-finetuned-ade-640-640"
    model = SegformerForSemanticSegmentation.from_pretrained(model_name)
    in_channels = model.decode_head.classifier.in_channels
    model.decode_head.classifier = nn.Conv2d(in_channels, num_class, 1, 1)
    model.to(device)
    optimizer = torch.optim.AdamW(params=model.parameters(), lr=lr, weight_decay=1e-4)
    criterion = smp.losses.DiceLoss(mode="multiclass", classes=num_class)
    lr_schedule = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=epochs,
eta_min=1e-6)
    mIoU_metric = MulticlassJaccardIndex(num_classes=num_class).to(device)
    dice_metric = Dice(num_classes=num_class, average="macro").to(device)

    if os.path.isdir(checkpoint):
        shutil.rmtree(checkpoint)

    os.makedirs(checkpoint)

    if not os.path.isdir(trained_model):
        os.makedirs(trained_model)

    writer = SummaryWriter(checkpoint)
    iteration_number = len(train_loader)
    best_dice = -1

    train_loss_list = []
    test_loss_list = []
    train_loss_meter = AverageMeter()
    mIoU_meter = AverageMeter()
    dice_meter = AverageMeter()
    no_improvement_counter = 0
    patience = 10

    for epoch in range(epochs):
        model.train()
        progress_bar = tqdm(train_loader, colour="cyan")
```

```
    train_loss_meter.reset()
    mIoU_meter.reset()
    dice_meter.reset()

    for iter, (image, mask) in enumerate(progress_bar):
        image, mask = image.to(device), mask.to(device)

        output = model(image).logits
        output = upsample(output)
        optimizer.zero_grad()
        loss = criterion(output, mask)
        loss.backward()
        optimizer.step()

        train_loss_meter.update(loss.item())
        progress_bar.set_description("Epoch {}/{} | Train/Loss {:0.4f} | lr {}".format(epoch + 1,
epochs, train_loss_meter.avg,
                                                                optimizer.param_groups[0]['lr']))
        writer.add_scalar("Train/Loss", train_loss_meter.avg, epoch * iteration_number + iter)

    lr_schedule.step()

    model.eval()
    progress_bar = tqdm(test_loader, colour="yellow")
    all_losses = []
    for image, mask in progress_bar:
        image, mask = image.to(device), mask.to(device)
        with torch.no_grad():
            output = model(image).logits
            output = upsample(output)

            mIoU = mIoU_metric(output, mask)
            dice = dice_metric(output, mask)

            loss = criterion(output, mask)
            all_losses.append(loss.item())
            mean_loss = np.mean(all_losses)

            mIoU_meter.update(mIoU.item())
            dice_meter.update(dice.item())

            progress_bar.set_description("Val | Loss {:0.4f} | mIoU: {:0.4f} | dice:
{:0.4f}".format(mean_loss, mIoU.item(), dice.item()))

    mean_loss = np.mean(all_losses)

    print("Val | Loss {:0.4f} | mIoU: {:0.4f} | dice: {:0.4f}".format(mean_loss, mIoU_meter.avg,
```

```
dice_meter.avg))

        writer.add_scalar("Val/mIoU", mIoU, epoch)
        writer.add_scalar("Val/dice", dice_meter.avg, epoch)

        checkpoint = {
            "model_state_dict": model.state_dict(),
            "optimizer_state_dict": optimizer.state_dict(),
            "epoch": epoch + 1,
            "dice": dice_meter.avg
        }

        torch.save(checkpoint, os.path.join(trained_model, "last.pt"))

        if best_dice < dice_meter.avg:
            best_dice = dice_meter.avg
            torch.save(checkpoint, os.path.join(trained_model, "best.pt"))
            no_improvement_counter = 0
        else:
            no_improvement_counter += 1

        if no_improvement_counter > patience:
            print("Early stopping!")
            break
        train_loss_list.append(train_loss_meter.avg)
        test_loss_list.append(mean_loss)

    plot_segmentation_metrics(train_loss_list, test_loss_list)
torch.cuda.empty_cache()
criterion = torch.nn.CrossEntropyLoss(ignore_index=255)
criterion = torch.nn.CrossEntropyLoss()
criterion = torch.nn.CrossEntropyLoss(ignore_index=255)

mask = torch.as_tensor(np.array(mask), dtype=torch.long)

import os
import torch
from torchvision.datasets import VOCSegmentation
from torch.utils.data import DataLoader
import torchvision.transforms as T
from torchvision.transforms import functional as TF
import numpy as np
from tqdm import tqdm  # ✅ Optional for progress bar

def train(num_class, batch_size, lr, epochs, model_save_path):
    # Dataset root
    voc_root = "./data"
```

```
    # Custom transform
    def voc_transform(image, mask):
        image = TF.resize(image, (256, 256))
        mask = TF.resize(mask, (256, 256), interpolation=T.InterpolationMode.NEAREST)
        image = TF.to_tensor(image)
        mask = torch.as_tensor(np.array(mask), dtype=torch.long)
        return image, mask

    # Load datasets
    train_dataset = VOCSegmentation(root=voc_root, year="2012", image_set="train",
download=True, transforms=voc_transform)
    val_dataset = VOCSegmentation(root=voc_root, year="2012", image_set="val", download=True,
transforms=voc_transform)

    # Data loaders
    train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
    val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)

    # Load model (updated syntax for weights)
    model = torch.hub.load("pytorch/vision:v0.10.0", "fcn_resnet50", weights=None,
num_classes=num_class)
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model = model.to(device)

    # Loss and optimizer
    criterion = torch.nn.CrossEntropyLoss(ignore_index=255)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)

    # Training loop
    for epoch in range(epochs):
        model.train()
        total_loss = 0
        loop = tqdm(train_loader, desc=f"Epoch [{epoch+1}/{epochs}]", leave=False)

        for images, masks in loop:
            images, masks = images.to(device), masks.to(device)
            outputs = model(images)['out']
            loss = criterion(outputs, masks)

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            total_loss += loss.item()
            loop.set_postfix(loss=loss.item())
```

```
        print(f" Epoch {epoch+1}/{epochs} completed - Total Loss: {total_loss:.4f}")

    # Save model
    os.makedirs(model_save_path, exist_ok=True)
    save_path = os.path.join(model_save_path, "fcn_resnet50.pth")
    torch.save(model.state_dict(), save_path)
    print(f" Model saved to {save_path}")

# === Run Training ===
epochs = 5          # Reduce for testing
lr = 1e-4
batch_size = 8      # Safe size for Kaggle/CPU
num_class = 21
trained_model = "/kaggle/working/trained_model"

train(num_class, batch_size, lr, epochs, trained_model)
def pred_show_image_grid(model_pth, device, test_dataset, num_classes, idx, num_imgs):
    # Load model checkpoint
    checkpoint = torch.load(model_pth)

    # Initialize and load model state
    model_name = "nvidia/segformer-b5-finetuned-ade-640-640"
    model = SegformerForSemanticSegmentation.from_pretrained(model_name)
    in_channels = model.decode_head.classifier.in_channels
    model.decode_head.classifier = nn.Conv2d(in_channels, num_class, 1, 1)
    model.to(device)

    model.load_state_dict(checkpoint['model_state_dict'])
    model.eval()
    # Load validation dataset

    images, orig_masks, pred_masks = [], [], []

    for i in idx:
        ori_img, orig_mask = test_dataset[i]
        img = ori_img

        # Predict mask
        img = img.float().to(device).unsqueeze(0)
        # (B, 21, H, W) -> (B, 1, H, W) -> (B, H, W)
        pred_mask = upsample(model(img).logits).argmax(dim=1).squeeze().cpu().numpy()

        # Colorize masks
        color_pred_mask = np.zeros((*pred_mask.shape, 3))
        color_mask = np.zeros((*orig_mask.shape, 3))

        for i, color in enumerate(VOC_COLORMAP):
```

```
            color_pred_mask[pred_mask == i] = np.array(color)
            color_mask[orig_mask == i] = np.array(color)

        # Store original image
        images.append(unorm(ori_img).permute(1, 2, 0))
        orig_masks.append(color_mask)
        pred_masks.append(color_pred_mask)

    # Combine images and masks for display
    images.extend(orig_masks)
    images.extend(pred_masks)

    # Plot images and masks
    fig = plt.figure(figsize=(20, 10))
    for i in range(1, 3*num_imgs + 1):
        fig.add_subplot(3, num_imgs, i)
        plt.imshow(images[i - 1])
    plt.show()
import os
import torch
import torch.nn.functional as F
import matplotlib.pyplot as plt
import numpy as np
from torchvision.datasets import VOCSegmentation
import torchvision.transforms.functional as TF
import torchvision.transforms as T
from torchvision import models

# 1. Set folder path
model_dir = "/kaggle/working/trained_model"

# 2. Check if directory exists
if not os.path.isdir(model_dir):
    raise FileNotFoundError(f"❌ The folder '{model_dir}' does not exist. Please check the model
save path.")

# 3. Look for .pt or .pth model file
files = os.listdir(model_dir)
model_file = next((f for f in files if f.endswith(('.pth', '.pt'))), None)
if model_file is None:
    raise FileNotFoundError(f"❌ No .pt or .pth model file found in the folder: {model_dir}")

# 4. Full model path
model_pth = os.path.join(model_dir, model_file)
print(f"✅ Using model file: {model_pth}")

# 5. Define VOC classes
```

```
VOC_CLASSES = [
    'background', 'aeroplane', 'bicycle', 'bird', 'boat',
    'bottle', 'bus', 'car', 'cat', 'chair',
    'cow', 'diningtable', 'dog', 'horse', 'motorbike',
    'person', 'pottedplant', 'sheep', 'sofa', 'train', 'tvmonitor'
]
num_classes = len(VOC_CLASSES)

# 6. Load test dataset
test_dataset = VOCSegmentation(
    root="./data", year="2012", image_set="val", download=True,
    transforms=lambda img, mask: (
        TF.to_tensor(TF.resize(img, (256, 256))),
        torch.as_tensor(np.array(TF.resize(mask, (256, 256),
interpolation=T.InterpolationMode.NEAREST)), dtype=torch.long)
    )
)

# 7. Visualization function
def pred_show_image_grid(model_pth, device, dataset, num_classes, idx_list, num_imgs=10):
    # Load model
    model = models.segmentation.fcn_resnet50(pretrained=False, num_classes=num_classes)
    model.load_state_dict(torch.load(model_pth, map_location=device))
    model.to(device)
    model.eval()

    # Plot predictions
    plt.figure(figsize=(15, num_imgs * 3))
    for i, idx in enumerate(idx_list[:num_imgs]):
        image, mask = dataset[idx]
        with torch.no_grad():
            output = model(image.unsqueeze(0).to(device))["out"]
            pred = torch.argmax(output.squeeze(), dim=0).cpu().numpy()

        # Show image, ground truth, prediction
        plt.subplot(num_imgs, 3, 3*i+1)
        plt.imshow(image.permute(1, 2, 0))
        plt.title("Image")
        plt.axis("off")

        plt.subplot(num_imgs, 3, 3*i+2)
        plt.imshow(mask)
        plt.title("Ground Truth")
        plt.axis("off")

        plt.subplot(num_imgs, 3, 3*i+3)
        plt.imshow(pred)
```

```
        plt.title("Prediction")
        plt.axis("off")
    plt.tight_layout()
    plt.show()

# 8. Predict and show
device = "cuda" if torch.cuda.is_available() else "cpu"
indexs = [10, 50, 90, 100, 150, 200, 250, 300, 350, 400]
pred_show_image_grid(model_pth, device, test_dataset, num_classes, indexs, num_imgs=10)

print("\n")
indexs = [1231, 772, 48, 368, 1144, 1443, 700, 610, 511, 753]
print("indexs =", indexs)
print("\n")

#Show images
pred_show_image_grid(model_pth, device, test_dataset, num_classes, indexs, num_imgs= 10)
print("\n")
indexs = [58, 89, 25, 32, 4, 64, 88, 85, 28, 38]
print("indexs =", indexs)
print("\n")

#Show images
pred_show_image_grid(model_pth, device, test_dataset, num_classes, indexs, num_imgs= 10)
```

CHAPTER-V

SYSTEM TESTING

5.1 Testing

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

Types of Tests

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions: identified functions must been exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

SYSTEM TEST

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as

specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

5.2 Test Strategy and Approach

Field testing will be performed manually and functional tests will be written in detail.

Test Objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features To Be Tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g., components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER-VI

CONCLUSION & FUTURE ENCHANCEMENTS

6.1 Conclusion

In this project, we demonstrated how semantic segmentation powered by deep learning can be effectively used for detailed image interpretation. By assigning a class label to every pixel in an image, the system provides a far more granular understanding compared to traditional classification or object detection methods.

The use of advanced architectures like U-Net and DeepLabV3+, trained on benchmark datasets such as PASCAL VOC and Cityscapes, has shown promising results in segmenting complex scenes involving multiple objects. The segmentation outputs enable rich insights for various real-world applications including autonomous driving, medical imaging, and smart surveillance.

The modular architecture and pixel-level precision make this system not only accurate but also scalable and adaptable to multiple domains. The ultimate aim of the project is to bridge the gap between raw image data and meaningful visual interpretation through the power of AI.

Future Enhancements:

1.Real-time Video Segmentation

Currently, the system is trained and evaluated on images and offline video frames. Future versions can implement real-time semantic segmentation to support live analysis in autonomous systems and surveillance networks.

2.Multi-class Object Recognition with Instance Segmentation

Enhancing the system to perform instance segmentation (distinguishing multiple objects of the same class) will improve its application in crowded or overlapping scenes such as traffic monitoring and crowd analysis.

3.Model Optimization for Edge Devices

Optimizing the trained models to run on low-resource devices (e.g., Raspberry Pi, Jetson Nano) will make the system suitable for edge deployment, especially in environments where cloud connectivity is limited.

4.Integration with GIS or Drone Data

Semantic segmentation can be extended to work with satellite or drone imagery for **agriculture**,

disaster response, and urban planning applications.

5.Domain Adaptation and Transfer Learning

Implementing transfer learning and domain adaptation techniques will allow the model to generalize better across new datasets with minimal retraining, enabling quick adaptation to new environments and tasks.

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